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# A COMPARATIVE ANALYSIS OF LAYOFF ANNOUNCEMENTS AND STOCK PRICE REACTIONS IN THE UNITED STATES AND JAPAN

PEGGY M. LEE*
*Roberto C. Goizueta Business School, Emory University, Atlanta, Georgia, U.S.A.*

*Downsizing and layoffs are an important mechanism for U.S. firms to cope with their strategic and economic environment. In contrast, the Japanese tradition of lifetime employment limits the ability of firms to employ layoffs as a strategic measure, relegating its use to conditions of financial distress. This paper provides the first comparison of layoffs in Japan and the United States and examines stock price reactions to layoff announcements in each country from 1990 to 1994. Agency theory and Aoki's cooperative game theory are employed to discuss differences in the governance structures of U.S. and Japanese firms and their implication for stock price reactions. Results show that layoff announcements trigger negative returns for both U.S. and Japanese firms. Specifically, layoff announcements of U.S. firms are associated with a negative 1.78 percent abnormal return, while layoff announcements for Japanese firms are associated with a negative 0.56 percent abnormal return. To better understand the impact of layoffs, this study examines the relationships between stock price reactions and various layoff characteristics (such as whether the layoff is proactive or reactive or whether the layoff is the first in the industry). Implications of the findings are discussed.* © 1997 John Wiley & Sons, Ltd.

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> The touchstone of the early 1990s was corporate downsizing—layoffs, restructuring, recapitalizing, bankruptcy, and emerging from bankruptcy.
> *Wall Street Journal* (16 March 1994)

Between 1987 and 1991, 85 percent of the *Fortune* 1000 firms in the United States downsized, affecting more than five million jobs (Cascio, 1993). This trend continued with 3100 layoffs per day in 1994 (Hitt *et al.*, 1994), and over 400,000 lost jobs in 1995 (Labich, 1996). Even Japanese firms widely known for lifetime employment policies have followed suit (Kang and Shivdasani, 1997). For example, NTT, Japan's telecommunications giant, announced plans to shed 45,000 jobs, and Nissan plans to lay off 7000 employees (*The Economist*, 10 February 1996: 25). Clearly, layoffs have become an important mechanism by which U.S. and Japanese firms cope with their competitive environment.

The relationship between layoffs and firm performance is a central concern for both practitioners and academics. Firms lay off workers for a variety of reasons. Layoffs may be a firm's way of optimally responding to demand shifts, to changes in its competitive position within the industry, or to financial distress. Moreover, layoffs are often a part of a firm's downsizing, downscoping, or restructuring strategies. Anticipated economic benefits from layoffs include lower expense ratios, enhanced profits, increased return-on-investment, and higher stock prices (Cascio, 1993).

Layoffs may also signal financial distress. In particular, firms with a stated lifetime employment policy are limited in their ability to use

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* Correspondence to: Peggy M. Lee, Roberto C. Goizueta Business School, Emory University, Atlanta, GA 30322-2710, U.S.A.

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layoffs as a strategic measure, relegating its use to conditions of financial distress. Worrell, Davidson, and Sharma (1991) place layoffs in two categories: financial distress, and restructuring/consolidation. They document that from 1979 to 1987 the net effect of layoff announcements on stock prices is negative.

This paper provides the first comparison of layoffs in the United States and Japan, and examines stock price reactions to layoff announcements of U.S. and Japanese firms from 1990 to 1994. This comparison is particularly interesting because the impact and management of layoffs may differ depending on the way firms resolve stakeholder relationships and problems in cost structure. Specifically, differences between U.S. and Japanese employment contracts may be reflected in the way layoffs are implemented and perceived by investors. This paper employs agency theory and Aoki's (1984) cooperative game theory to discuss these governance and organizational differences.

To better understand the impact of layoffs, I examine stock price responses to differing layoff characteristics. Stock price reactions may differ depending on whether layoffs are permanent or temporary, small or large, a single announcement or part of a series of announcements, and proactive or reactive. Also, the sequencing of the layoff—whether the layoff announcement is the first in a given industry or a subsequent announcement—may affect stock prices. These distinctions enhance our understanding of the effectiveness of layoffs. Accordingly, hypotheses detail the expected differences in abnormal returns of layoff announcements for U.S. and Japanese firms, across various layoff characteristics.

## DEFINITIONS OF LAYOFFS AND RELATED STRATEGIES

Firms employ layoffs as a means of and as a part of strategic responses to the competitive environment. Layoffs often accompany downsizing, downscoping, or restructuring strategies. Downsizing refers to a 'deliberate decision to reduce the work force that is intended to improve organizational performance' (Kozlowski *et al.*, 1993: 267). In such cases, laying off workers may be employed as the sole measure—unaccompanied by additional restructuring strategies—for improving firm performance. Firms employing only layoffs are often reacting to poor performance or to the competitive environment.

However, layoffs may be part of a broader strategic initiative, such as a planned, well-considered downscoping strategy. Hoskisson and Hitt define downscoping as 'the process through which firms reemphasize their core businesses' (1994: 164). Thus, one distinction between layoffs is whether they are *proactive*, part of deliberate downscoping or restructuring strategies or *reactive*, employed as the sole response to financial distress.[1] Proactive layoffs are purported to maintain competitiveness or to correct performance downturns before they become severe (Byrne, 1994; Hughes and Clark, 1988), while reactive layoffs are efforts to turnaround firms with poor or declining performance (Worrell *et al.*, 1991).

### Layoffs in Japan

The governance structures and employment policies of Japanese firms differ from their U.S. counterparts. For example, main banks hold both debt and equity in Japanese firms, whereas U.S. banks are not permitted to own equity (except under Chapter 11 conditions). Further, equity crossholdings between firms play an important role in Japan but not in the United States (Hoshi, Kashyap, and Scharfstein, 1990; Kaplan, 1994). Despite these differences, Japanese firms also undergo substantial reorganizational activities subsequent to poor performance. These activities, which include layoffs, are influenced by the presence of main banks and other large shareholders (Kang and Shivdasani, 1997). Thus, differences in governance structures and employment policies may affect the use and impact of layoffs in Japan and the United States. The following section provides a theoretical development of how governance structures may affect employment policies and investors' perceptions of layoffs.

[1] The distinction between proactive and reactive strategic actions is supported in Hrebiniak and Joyce's (1985) theoretical model of the causes of strategic actions.

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## THEORETICAL PERSPECTIVES ON LAYOFFS

### Agency and contracting theory

Agency theory contends that there exists a principal–agent or moral–hazard problem between the owner (the principal) and the employee (the agent) (Holstrom, 1979). Firms typically manage this moral hazard problem in three ways.

First, principals monitor workers' efforts and outcomes. Monitoring of worker efforts and outcomes is often visible through approval requirements and performance appraisals. These features are common to both U.S. and Japanese firms.

Second, *ex post* settling up (or the threat of *ex post* settling up) may be used to mitigate the agency problem *ex ante*. If layoffs are performance-based, then layoffs and the threat of further layoffs reinforces this *ex post* settling up mechanism. However, large layoffs may also result from agency problems between top management and shareholders, rather than from agency problems between managers and employees. In this scenario, layoffs are used to reverse unwise investment decisions (e.g., AT&T's acquisition of NCR). In both cases, layoffs may not provide a complete solution to the performance problem. For example, firms using layoffs to mitigate agency problems may suffer from competitive problems due to poor productivity. Alternatively, by waiting until productivity problems arise, these firms may become less competitive.

Layoffs may actually exacerbate poor productivity with remaining workers. In fact, survivors of layoffs tend to experience psychological stress, job insecurity, and anger (Brockner, 1988), and their commitment decreases when they identify with victims of layoffs (Brockner *et al*., 1986). Lower commitment may result in slower decision-making, decreased productivity and communication problems. Cascio (1993) argues that few of the cost-cutting, profit-maximizing goals of layoffs have actually been realized. Survey and other results suggest that the benefits that firms and shareholders hope for often fail to develop. In a 1993 survey by Wyatt and Company, less than one-half of the 1005 firms surveyed reported that profits actually increased as much as expected after layoffs (Hitt *et al*., 1994). Worrell *et al*. (1991) find that layoff announcements over the 1979–87 period trigger a small but significant negative stock price reaction. This evidence suggests that layoffs have been only somewhat successful.

Third, managers may put pay levels at risk. For example, firms use performance-based contracts as a method of ensuring a certain level of effort and competence. A firm may be considered 'a nexus of long-term contracts between shareholders and stakeholders' (Shleifer and Summers, 1988). The primary reason for long-term contracts is to promote relationship-specific capital investments by stakeholders. Employees expend effort to complete tasks well, only when there is a subsequent reward.

Using the analogy of a firm as a nexus of contracts, Aoki (1989) argues that agency problems occur because gaps in the contract exist. U.S. firms are viewed as a nexus of *explicit* contracts (Jensen and Meckling, 1976) which use both *ex post* settling up mechanisms and *ex ante* contracts to fill this gap. In contrast, large Japanese firms may be viewed as a nexus of *implicit* contracts, aligning interests primarily through *ex ante* contracts, such as an implicit lifetime employment guarantee and a wage system based on seniority, base, and merit (Ouchi and Price, 1978). However, the distinction between explicit and implicit contracts is not always clear. For U.S. firms, early retirement and severance packages are examples of fulfilling quasi-contracts and may be regarded as partial payment for breach of an employment contract.

Perhaps the sharpest distinction between U.S. and Japanese firms is the implicit guarantee of lifetime employment for permanent Japanese workers.[2] In Japan, lifetime employment contracts encourage high levels of individual effort and cooperation between employees (Garvey and Swan, 1992). A violation of this implicit contract is often greeted with hostility (Ouchi and Price, 1978), perhaps not unlike the stress and decreased commitment that layoff survivors in U.S. firms face (Brockner, 1988). Thus, empirical evidence and agency theory arguments suggest that layoff announcements signal a performance problem. Therefore:

*Hypothesis 1: The stock market reactions of*

[2] It is important to note that temporary workers and those employed in smaller Japanese companies are generally not given the same lifetime employment guarantee.

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*layoff announcements will be negative in the United States and Japan.*

Historically, layoffs have resulted in the longest, and most bitter strikes in postwar Japan, (Kenney and Florida, 1989). Consequently, guaranteed employment has become a deeply embedded feature of many large Japanese firms. One oftencited reason for Japan's success is the ability to maintain valuable implicit agreements with employees (Kester, 1991); one valuable agreement is lifetime employment for a large set of permanent workers (Garvey and Swan, 1992).[3] Since layoffs are not common in Japan, large Japanese firms view workers as cooperative partners. Thus, Aoki's cooperative game theory of the firm may be an alternative way to examine the employer–employee relationship in Japanese firms.

## Aoki's cooperative game theory of the firm

Aoki (1984) argues that both neoclassical economic and agency theories do not treat the interactions among shareholders, managers, and employees. In his cooperative game theory, the firm is viewed as a coalition with shareholders and employees as its members; labor decisions airse from the balance of power between members of the firm.

Aoki's cooperative game theory has been applied to the Anglo-American unionized firm and the Japanese firm. Japan's paternalistic labor policy stresses the quasi-family relationship with employees and lifetime employment, and is consistent with the collective bargaining of Aoki's model. In contrast, the labor market turnover in Anglo-American unionized firms emphasizes conflict and is consistent with neoclassical theory. Both agency theory and Aoki's theory suggest that U.S. and Japanese firms differ fundamentally in their use of layoffs.

Consistent with the analogy of a firm as a coalition of shareholders and employees, Japanese firms are characterized by a clanlike structure, a strong corporate culture of 'familism' and a guarantee of lifetime employment (Ouchi and Price, 1978; Ouchi, 1980; Clark, 1988). Clanlike structures are often criticized for imposing social costs, such as employment welfare and community responsibility which are essential to maintaining a strong corporate culture. Deviation from established standards, including those that may be important for adaptation and survival, is usually greeted with hostility (Ouchi and Price, 1978; Ouchi, 1980). Layoffs in Japan are not as common, as perfunctory, or as easily accomplished as in the United States.

The aversion to layoffs in Japan has resulted in creative forms of restructuring and cost-cutting. Some firms absorb the impact of lower revenue by cutting employee bonuses, reducing overtime, hiring fewer graduates, making fewer products, and slashing capital spending (Kang and Shivdasani, 1997). Other firms have chosen to retain excess employees despite the economic advantages of laying off these workers.

In contrast, layoffs are a common phenomenon in the United States; anecdotally, major downsizings have resulted in positive stock market effects. For example, recent layoff announcements at AT&T have elicited positive responses from the stock market, perhaps signaling that the firm is serious about dealing with its problems. Moreover, the example of General Dynamics illustrates that there are economic benefits, such as increased rates of return realized by shareholders, to downsizing (Dial and Murphy, 1995). Given these arguments:

*Hypothesis 2a: The stock market reactions of layoff announcements in Japan will be more negative than the stock market reactions of layoff announcements in the United States.*

Another important feature of the Japanese firm and of Aoki's theory is the cooperative role that shareholders play. Large cross-holdings are common among Japanese firms, particularly those part of a *Keiretsu* (Hoshi *et al*., 1990; Kester, 1991), and may affect the impact of layoffs on stock prices. Table 1 presents information on the ownership structure for a large sample of Japanese firms and the annual portfolio turnover rates of various security owners.

[3] Large Japanese firms employ a much smaller group of temporary workers that change over time and provide a buffer and flexibility to respond to the competitive conditions in the environment. This group of temporary workers is only a small but growing percentage of the overall workforce. With the recession in Japan, temporary and part-time workers have increased to subsume a quarter of the work force (*The Economist*, 10 February 1996). This is, however, a recent occurrence that is highly discouraged by Japan's Ministry of Labor.

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Table 1. Ownership structure and portfolio turnover rates for Japanese firms

| | Percentage ownership | Annual turnover (%) |
|---|---|---|
| Banks | 20.5 | 39.2 |
| Investment trusts | 1.8 | 189 |
| Insurance companies | 17.7 | 2.3 |
| Business corporations | 24.5 | 21.2 |
| Individuals and others | 23.9 | 106 |
| Foreigners | 4.7 | 103 |
| Others | 6.9 | N/A |

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Table 1 shows that large proportions of the stock are held by banks and insurance companies who trade infrequently, particularly relative to the high portfolio turnover rates of individuals and foreigners. If these investors are 'informed' and are aware of impending layoffs prior to their announcements, they may sell the stock prior to the announcement (i.e., there may be leakage effects). Alternatively, Japanese firms may simply be financed by long-term investors who are not inclined to react to the announcement. A typical listed Japanese firm has about 70 percent of its shares held by other corporations, with which it has a reciprocal shareholder relationship (Sheard, 1994). As such, "most corporate shareholdings in Japan are based on what Japanese businessmen term stable shareholding arrangements" (Sheard, 1994: 314). These interlocking shareholdings are another feature of Japan's patient capital structure and may result in a relatively small price effect.

The rise in institutional ownership of publicly held firms may provide a source of patient capital for U.S. firms. Faced with a declining performance of companies in which they own stock, institutions can either attempt to influence the way a firm is governed (in hopes of improved performance) or simply sell their shares. The former is a 'patient' response in the sense that institutions retain their ownership stake. Some evidence suggests that institutional investors in U.S. firms may opt for the 'patient' route. For example, institutional investors vote more actively on antitakeover amendments (Brickley, Lease, and Smith, 1988), take actions affecting corporate governance but do not sell their stock (Wahal, 1996), and may increase innovations, which require a long-term orientation (Kochhar and David, 1996). On the other hand, Pound (1988) finds that institutions do not vote against management and Lang and McNichols (1993) show that most categories of institutional investors are not patient. Given this conflicting evidence, institutions, on average, are an unlikely source of patient capital for U.S. firms. Thus, a competing hypothesis is:

*Hypothesis 2b: The stock market reactions of layoff announcements in the United States will be more negative than the stock market reactions of layoff announcements in Japan.*

In addition to comparing the stock market reactions to layoff announcements in the United States and Japan, I examine the stock price effects of differing layoff characteristics. Distinguishing layoffs based on various characteristics allows for a fuller picture of the impact of layoff announcements on stock prices. A model of hypothesized relationships is presented in Figure 1 and discussed below.

## Downsizing characteristics: hypotheses

Layoffs can be viewed as forms of corporate signaling (Asquith and Mullins, 1986), impression management (cf. Staw, McKechie, and Puffer, 1983), or institutional or mimetic isomorphism (Norman, 1995). The nature of the announced layoff—whether the layoff is permanent of temporary, a single announcement or part of a series of announcements, small or large, and proactive or reactive—sends signals to investors about the firm's condition. Thus, differing layoff characteristics are expected to trigger disparate stock market reactions.

### *Permanent vs. temporary layoffs*

Permanent layoffs are due to long-term changes in the firm and its competitive environment, while temporary layoffs represent alternatives to inventory accumulation, simply a convenient way of dealing with seasonal demand shifts

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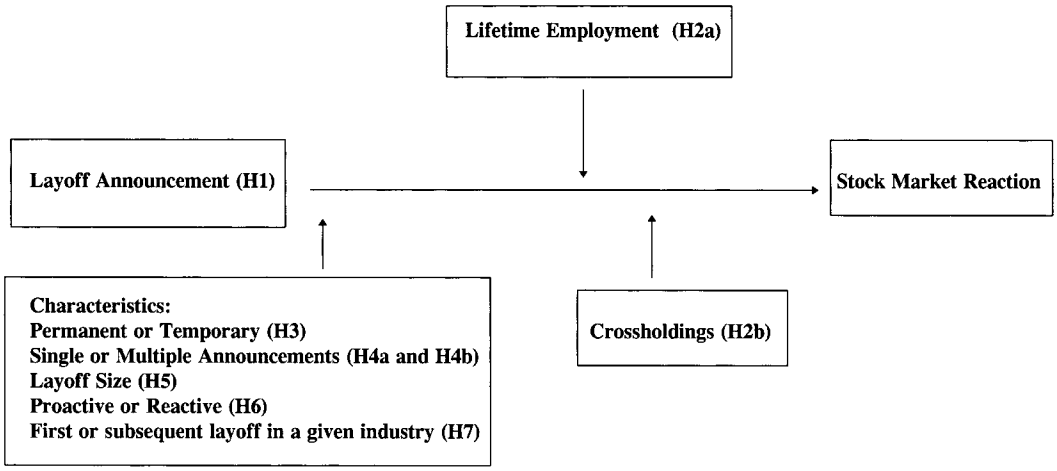



Figure 1. Model of hypothesized relationships between layoffs and stock market prices

(Saffer,1983).[4] Thus, permanent layoffs are assumed to be long-term changes that impact a firm more significantly than temporary disruptions in the workforce. Temporary layoffs may signal a firm's expectation of a quick positive turnaround, while permanent layoffs may signal the need for long-term measures to achieve competitive advantage or parity.

The signals contained in announcements of permanent and temporary layoffs may also extend to Japanese firms. Japanese firms provide a relatively stable employment for a core of 'permanent workers,' while maintaining a smaller set of temporary employees that changes over time (Garvey and Swan, 1992). This smaller set of temporary workers gives Japanese firms more flexibility to deal with environmental changes. Laying off permanent workers in Japan unmistakably sends a signal about the expected duration of the firm's condition. Hence:

*Hypothesis 3: Stock prices will react more negatively if layoffs are permanent rather than temporary for both U.S. and Japanese firms.*

[4] Interestingly, Greer and Ireland (1992) describe a contrarian human resources strategy, whereby firms derive benefits from countercyclical hiring. Firms hire high-quality employees that become available during downturns. Conceivably, some firms may simultaneously lay off low-quality workers while hiring high-quality workers. However, at the time of a layoff announcement, firms are unlikely to reveal plans for a contrarian hiring policy. Concerns about equity in layoff processes (Brockner *et al*., 1986) are well known, and the acknowledged use of a worker substitution policy may exacerbate these concerns.

*Multiple vs. single announcements*

Layoffs may be announced as a single event or as part of a series of restructuring efforts (i.e., a series of *different* layoff announcements). Multiple announcements may signal that the firm's earlier layoffs were insufficient to generate a turnaround in the firm's performance. For example, Kodak has undergone numerous restructurings since 1982, and is still attempting to improve performance (Hoskisson and Hitt, 1994). These restructuring efforts included numerous layoffs, with four announcements between 1993 and 1995. Similarly, Honeywell, Xerox, and IBM have announced multiple cutbacks (Cascio, 1993). These examples suggest that the first layoff was insufficient to improve performance, and these firms often announce subsequent layoffs for further restrcturing. Muiltiple announcements may signal that despite restructuring efforts, firms have been unable to recover from competitive difficulties. Hence:

*Hypothesis 4a: Stock prices will react more negatively if the layoff is one of several layoff announcements (i.e., multiple announcements) rather than if the layoff is a single event for both U.S. and Japanese firms.*

On the other hand, the market's reaction to the first announcement should be related to its expectations of the impact of the layoffs on the firm. Thus, single announcements may carry more information than multiple announcements. The first announcement reflects an unbiased forecast

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of the layoff's impact and anticipated restructuring activity. Under this scenario, the marginal impact of subsequent announcements should be smaller. Hence:

*Hypothesis 4b: Stock prices will react more negatively if the layoff is a single event rather than part of multiple announcements for both U.S. and Japanese firms.*

### *Layoff size*

The percentage of the work force being laid off should also be related to the stock price reaction. Intuitively, the magnitude of a layoff conveys a signal about the severity of the firm's problems. Additionally, larger layoffs accompany additional costs including severance pay entitlements, high unemployment taxes, and extended health benefits. These costs are not unlike those generated in plant closings (Addison and Portugal, 1987). Thus:

*Hypothesis 5: Stock prices will react more negatively if the layoff affects a greater percentage of the firm's work force for both U.S. and Japanese firms.*

### *Proactive vs. reactive layoffs*

In addition to corporate signaling, firms may use layoff announcements as a form of impression management. Pfeffer and Salancik (1978) contend that firms are rational entities that need to justify actions and performance. One of management's major roles is to provide the context and logic behind managerial actions so that these actions can be 'properly' interpreted by investors; this justification is a form of impression management. Management needs to convey negative news in a manner which minimizes its impact (Staw *et al*., 1983). One manifestation of impression management is to make work force reductions appear proactive rather than reactive.

Generally, proactive layoffs are defined as those that are part of an overall strategic objective, while reative layoffs are responses to poor financial performance. If impression management is effective, stock prices should react differently to proactive vs. reactive layoff announcements. Specifically, since proactive layoffs are part of a broader strategic initiative, shareholders may perceive proactive layoffs better than reactive layoffs, suggesting that the firm may be slow to respond to its competitive environment. Furthermore, some have suggested that layoffs are an ineffective strategy if firms fail to make other strategic (i.e., moving out of a line of business) or structural (i.e., combining divisions, reducing layers of management, consolidating production facilities, etc.) changes. Impression management may further accentuate differences between proactive and reactive layoff announcements. Thus:

*Hypothesis 6: Stock prices will react negatively if the layoff is perceived as reactive rather than proactive for both the United States and Japan.*

### *Sequencing of layoff announcements within industry*

In the United States, a firm's decision to lay off workers may also be a result of institutional or mimetic isomorphism (Norman, 1995).[5] In other words, layoffs may have become 'institutionalized', as a common way of cutting down costs, improving efficiency, and *proactively* responding to the environment.

As a result, the sequencing of the layoffs within an industry may have an impact on the perception of investors. Firms within a given industry may feel pressure to mimetically follow the trend of layoffs if competing firms have also implemented the same cost-cutting measures. In other words, the first firm in an industry to announce a layoff during a specified time period may be subject to the scrutiny of investors—wondering if the firm is signaling financial distress. On the other hand, investors observing subsequent layoff announcements by other firms in the same industry are likely to perceive these announcements as nothing more than 'follow-the-leader' behavior (Haveman, 1993). Hence:

*Hypothesis 7: In the United States stock prices will react more negatively to the first layoff announcement in a given industry than to subsequent announcements within a given industry.*

[5] This relationship is not tested for the Japanese sample because layoffs remain a relatively uncommon phenomenon.

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## DATA AND METHODOLOGY

### Data sources and sample construction

A sample of layoff announcements for the United States has been obtained from the *Wall Street Journal* from 1990 to 1994. Layoff announcements for government organizations and private companies have been deleted from the sample. Twenty-six layoff announcements with confounding events (i.e., CEO dismissal, new executive appointments, mergers, etc.) have also been deleted. This sampling procedure has resulted in 300 unique layoff announcements for firms listed on the New York Stock Exchange, American Stock Exchange and NASDAQ. Since all three exchanges are represented, the sample is not biased toward small or large firms.

The Japanese layoff announcement dates over the same period have been obtained from the *Asian Wall Street Journal*, *Nikkei Weekly*, *Nihon Keizai Shinbun*, *South China Morning Post*, *Financial Times*, *Xinhua Newswire*, *Japanese Economic Newswire*, and the *International Herald Tribune* over the same time period. The *Japan Company Handbook* is used to identify whether the firm is traded on Section 1 or Section 2 of the Tokyo Stock Exchange. A total of 73 Japanese layoff announcements are identified. The smaller Japanese sample seems representative of the number of firms laying off workers over the period. Kang and Shivdasani (1997) provide corroborating evidence, reporting that between 1985 and 1990 only 14 of 87 poorly performing Japanese firms in their sample resorted to layoffs.

One potential problem with the Japanese announcement dates is that several of the sources are weekly. Thus, the announcement dates may not be identified with enough precision to capture the information relating to the layoff announcement in the abnormal returns. Several robustness checks have been conducted to assess the significance of this potential problem. First, event study returns have been estimated only for the subsample of Japanese layoff announcements whose dates are identified with precision from *daily* news sources.[6] Second, for the entire sample, cumulative abnormal returns are calculated over a −2, +2 window (returns for the U.S. sample are also calculated over a −2, +2 window to ensure comparability). Weekly Japanese publications are typically published on Wednesday, containing news from the previous Wednesday to the Tuesday prior to publication. In these cases, day 0 is identified as the middle of the 5-day period (not including weekends). Thus, a −2, +2 window covers the period of the announcement and captures the information contained in the layoff announcement.

To verify the validity of this procedure, a subsample of announcement dates which are reported in both daily and weekly news sources have been examined. In all cases, the announcement dates from daily sources are contained in the −2, +2 window. Brown and Warner (1985) suggest that when information release dates are identified with uncertainty, a wider window may be used. To the extent that a wider window introduces noise, returns are estimated with greater error, reducing the power of test statistics. However, a 5-day window does not introduce a significant amount of estimation error, thereby alleviating concerns regarding the power of test statistics.

For both the U.S. and Japanese samples, the news articles are the source for identifying whether the layoff is permanent or temporary, proactive or reactive, and whether the announcement is part of a series of layoffs or a single event.[7] Layoff announcements typically indicate the number of employees being laid off. Data on the total number of employees in U.S. firms, reported in the previous quarter, have been obtained from *Compact Disclosure* and are used to compute the percentage of the workforce laid off. Similarly, data on the number of employees in Japanese firms is obtained from *Worldscope*. The identification of most of the remaining layoff characteristics is relatively straightforward.

One exception is the identification of proactive vs. reactive announcements. Proactive announcements are defined as layoffs that are part of a strategy or a restructuring plan which anticipates

[6] The results for the subsample of Japanese firms with precisely identified announcement dates are −0.83 percent for the window (−1, +1), and −1.07 percent for the window (−2, +2). The sample contains 18 observations, and these returns are significant at the 0.01 level.

[7] The news sources are also used to identify any confounding events concurrent to the layoff announcement. Three events have been detected. All the results reported in the paper are qualitatively unchanged if these observations are removed from the sample.

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the direction of the competitive environment. Reactive announcements are layoffs that are a direct response to financial distress. To ensure the accuracy of the data, a coder reliability check has been conducted for both the U.S. and Japanese samples. In 82 percent of the cases, the secondary coder's data match the primary coder. Observations with discrepancies have been reexamined by a third coder. Agreement between two coders was required to categorize the layoff as either proactive or reactive.

Stock returns data for U.S. firms come from the tapes provided by the Center for Research in Securities Prices (CRSP) of the University of Chicago. Industries are identified by 4-digit SIC codes reported in the CRSP tapes. Stock price data on Japanese firms come from *Datastream International*. Since daily price data are not available for 15 of the Japanese firms, the Japanese layoff sample consists of 58 observations for most tests. The number of employees involved in the layoff are unavailable for 18 of these observations; some tests requiring this information are conducted on a subsample of 40 observations.

### Event study methodology

Standard event study methodology (Fama *et al*., 1969) is employed to measure abnormal returns for both the U.S. and Japanese firms. The use of stock price reactions in event studies is common in financial economics and increasingly employed in strategic management (e.g., Bettis and Weeks, 1987; Worrell *et al*., 1991; Bromiley and Marcus, 1989; Lubatkin, 1987). The premise behind event studies is that price changes reflect investor expectations of changes in a firm's future performance based on new information.

## RESULTS

Table 2 presents abnormal returns over the −2, +2 window for the sample of U.S. and Japanese layoff announcements. Table 2 shows that the 5-day cumulative return is a significant −1.78 percent for the full sample of U.S. firms, providing strong support for Hypothesis 1. These results are markedly different from the −0.4 percent 3-day cumulative return reported by Worrell *et al*., (1991) for a sample of 441 layoff announcements over an 8-year period from 1979 to 1987. Clearly, the magnitude of the stock price impact of layoff announcements is more severe in the 1990s.

Table 2 also shows that the 5-day cumulative abnormal return for the full sample of Japanese firms is a significant −0.56 percent, thereby supporting Hypothesis 1. This return is significantly less negative than the U.S. sample, rejecting Hypothesis 2a and supporting Hypothesis 2b ($t = 2.86$, $p < 0.01$).

The U.S. subsample of permanent and temporary layoffs shows that the abnormal return for permanent layoff announcements is −2.00 percent vs. an insignificant 1.28 percent for temporary layoffs. Abnormal returns to the announcement of permanent layoffs are significantly more negative than for temporary layoffs, supporting Hypothesis 3 ($t = 3.90$, $p < 0.01$). Of the 58 Japanese layoff announcements, 56 are permanent, precluding the possibility of an examination of Hypothesis 3.

The sample of single vs. multiple announcements allows for a test of Hypotheses 4a and 4b. Consistent with Hypothesis 4b, U.S. results show that single announcements are significantly more negative (−3.58%) than multiple announcements (−0.65%) ($t = 3.01$, $p < 0.01$). In Japan, multiple announcements are more negative than single announcement returns (−0.69% vs. −0.32%). Thus, the Japanese sample supports Hypothesis 4a while the U.S. sample supports Hypothesis 4b.

As a test of Hypothesis 5, the sample of U.S. layoffs has been broken into five quintiles based on the percentage of the work force laid off. The results, although not strong, provide support for Hypothesis 5. Hypothesis 5 suggests that there should be a monotonic relationship between the abnormal return and the percentage of work force laid off. While the largest quintile (5) has a much more negative return (−3.66%) than the smallest quintile (1) (−1.35%), two of the intermediate quintiles' (3 and 4) returns are lower than the return in quintile 1. Nonetheless, a chi-square test rejects the hypothesis that the returns for the five quintiles are equal with a *p*-value of 0.001; partial support is provided for Hypothesis 5. These results suggest a curvilinear relationship between the percentage of the work force laid off and the cumulative abnormal return.

Similarly, the sample of Japanese layoffs is broken into four quartiles based on the percentage of the work force laid off (the smaller sample precludes the finer quintile breakdown). None of

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Table 2. Abnormal returns of layoff announcements for U.S. and Japanese firms: 1990–94

| Sample and subsamples | Sample size | Event window (−2, +2) | *Z*-statistic |
|---|---|---|---|
| All U.S. firms | 300 | −1.78%*** | (−4.47) |
| All Japanese firms | 58 | −0.56%** | (−2.78) |
| U.S.: permanent layoffs | 271 | −2.00%*** | (−4.85) |
| U.S.: temporary layoffs | 18 | +1.28% | (1.26) |
| U.S.: single announcement | 115 | −3.58%*** | (−4.48) |
| U.S.: multiple announcements | 185 | −0.65%* | (−2.16) |
| Japan: single announcements | 19 | −0.32% | (−1.20) |
| Japan: multiple announcements | 31 | −0.69%* | (−2.48) |
| *Percentage of work force* | | | |
| U.S.: quintile 1 (< 0.017) | 53 | −1.35%* | (−2.56) |
| U.S.: quintile 2 (< 0.042) | 52 | −1.86%*** | (−3.36) |
| U.S.: quintile 3 (< 0.083) | 52 | −1.06% | (−0.85) |
| U.S.: quintile 4 (< 0.130) | 50 | −1.53% | (−1.49) |
| U.S.: quintile 5 (< 0.520) | 54 | −3.66%* | (−2.29) |
| *Percentage of work force* | | | |
| Japan: quartile 1 (< 0.035) | 8 | −0.73% | (−1.65) |
| Japan: quartile 2 (< 0.090) | 10 | −0.63% | (−0.97) |
| Japan: quartile 3 (< 0.160) | 11 | −0.47% | (−0.99) |
| Japan: quartile 4 (< 0.250) | 12 | +0.32% | (+1.14) |
| U.S.: proactive layoffs | 90 | −0.24% | (−1.57) |
| U.S.: reactive layoffs | 165 | −2.72%*** | (−5.85) |
| U.S.: first in industry | 92 | −3.59%*** | (−4.60) |
| U.S.: subsequent layoffs | 208 | −0.97%* | (−2.29) |

***$p < 0.001$; **$p < 0.01$; *$p < 0.05$
*Z*-statistics indicated in parentheses.

the quartile returns are statistically significant, perhaps because of the small sample. Thus, the Japanese sample does not appear to support Hypothesis 5.

The sample of proactive vs. reactive layoff announcements of U.S. firms also yields interesting and significantly different results. Proactive announcements have a smaller negative return (−0.24%); on the other hand, reactive announcements have a significantly negative return of −2.72 percent ($t = 2.35$, $p < 0.01$). These results provide support for Hypothesis 6. Unfortunately, all of the Japanese layoff announcements are reactive, precluding the possibility of any analysis of Hypothesis 6 for Japanese firms.

U.S. layoff announcements are also classified into the first layoff announcement in a given industry, or a subsequent layoff. Interestingly, results show that the first layoff announcement in a given industry is associated with a −3.59 percent return, while subsequent announcements are associated witha smaller −0.97 percent return. A standard *t*-test for the difference in these returns is significant at the 0.01 level ($t = 2.90$), providing support for Hypothesis 7.[8].

[8] In the early 1980s, layoffs were rare and considered a drastic measure. Recently, layoffs have been euphemistically referred to as 'rightsizing'. This change in firm, media, and investor perceptions of layoffs may be evident in the differing stock price reactions over the years; more recent layoff announcements may actually be applauded by investors. As a robustness check for Hypothesis 7, I also examine stock price reactions to layoff announcements for each year. Results show a statistically significant monotonic pattern from a −4.60 percent in 1990 to +1.29 percent in 1994. This evidence suggests

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### Multivariate regression and correlations: U.S. sample

Since the characteristics of layoffs may jointly influence stock price reactions, they have been further examined in a multivariate setting. A multivariate regression has been estimated with the cumulative abnormal returns as the dependent variable and layoff characteristics as independent variables. The results for the U.S. sample are presented in Table 3. The multiple layoff dummy and the proactive layoff dummy are statistically significant in a manner similar to the univariate results in Table 2; these results support Hypotheses 4b and 6. However, the permanent layoff and first-in-industry dummies are not significant; Hypotheses 3 and 7 are not supported. A continuous variable indicating the percentage of workforce laid off is used rather than quintile breaks since the continuous variable provides more information. This coefficient on this variable is negative but not significant. The square of the percentage of the work force laid off is also employed as an independent variable to assess the curvilinear nature of the relationship between the size of the layoff and the abnormal returns. This squared term is significant, providing additional support for the curvilinear relationship between the percentage of the work force laid off and abnormal returns. In general the multivariate analysis supports the inferences made from the univariate tests. In some cases, statistical significance is reduced but the sign of the abnormal returns across downsizing characteristics remains the same.

Several robustness checks have been performed on the reported regression to ensure the validity of the results. Dummy variables for quintile breaks are used instead of the continuous percentage of work force laid off variable. Consistent with the univariate results in Table 2, the percentage of the workforce laid off is negatively related to abnormal returns. The regression has also been estimated with interaction terms between the characteristics of the layoffs. These interaction terms are not significant. Since the dependent variable is estimated with error, weighted least squares (instead of OLS) has also been used. The standard error of the abnormal return is used as the weight. The result remains unchanged.

Finally, collinearity among the independent variables may result in econometric problems in the cross-sectional regression. Consequently, I compute Pearson correlation coefficients between the independent variables. Table 4 presents a correlation matrix for the U.S. sample. Table 4 shows that with the exception of two variables the correlations between the independent variables are small. There is a significant correlation between the first layoff in a given industry and multiple/single dummies. Examination of variance inflation factors in the cross-sectional regression shows, however, that this correlation does not substantially inflate the standard errors of the independent variables. The high correlation between the percentage of the work force laid off and its squared term does, however, inflate standard errors. Indeed in a regression without the squared term, the percentage of the work force laid off has a negative and significant coefficient.

Table 3. Cross-sectional regression of abnormal returns on layoff characteristics for U.S. firms

| Variables, $N = 223$ | Parameter estimates |
|---|---|
| Intercept | –0.02 |
| Permanent layoff dummy (*permanent is coded 1; temporary is coded 0)* | –0.01 |
| Multiple layoff dummy (*multiple is coded 1; single is coded 0)* | *0.02†* |
| Percentage of work force laid off (*continuous variable*) | –0.11 |
| Percentage of work force laid off—squared | –0.79* |
| Proactive dummy (*proactive is coded 0; reactive is coded 1*) | –0.02* |
| First in industry (*first is coded 1; subsequent announcements coded 0*) | –0.01 |
| $R^2$ | 0.09 |
| *F*-value | 4.27*** |

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $\dagger p < 0.1$

### Multivariate regression and correlations: Japanese sample

Table 5 presents a cross-sectional regression for the Japanese sample. Data on the percentage of

that investors' perceptions of layoffs are more positive, and may explain the positive stock market reaction to recent downsizings, such as AT&T.

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Table 4. Correlations of independent variables for U.S. firms

| Variables | Permanent or temporary | Multiple or single event | Percentage of work force | Proactive or reactive | First layoff in an industry | (Percentage of work force)$^2$ |
|---|---|---|---|---|---|---|
| Permanent or temporary | 1.00 | | | | | |
| Multiple or single event | 0.18** | 1.00 | | | | |
| Percentage of work force | –0.06 | –0.15** | 1.00 | | | |
| Proactive or reactive | 0.01 | 0.05 | 0.00 | 1.00 | | |
| First layoff in an industry | –0.14* | –0.52*** | 0.09 | –0.09 | 1.00 | |
| (Percentage of work force)$^2$ | –0.02 | –0.09 | 0.90*** | 0.02 | 0.16** | 1.00 |

***$p < 0.001$;; **$p < 0.01$; *$p < 0.05$

Table 5. Cross-sectional regression of abnormal returns on layoff characteristics for Japanese firms

| Variables, $N = 58$ | Parameter estimates |
|---|---|
| Intercept | –0.00 |
| Multiple layoff dummy (*multiple is coded 1; single is coded 0*) | –0.01 |
| Percentage of work force laid off (*continuous variable*) | 0.01 |
| Proactive dummy (*proactive is coded 0; reactive is coded 1*) | 0.16 |
| $R^2$ | 0.04 |
| *F*-value | 0.56 |

the work force laid off are unavailable for 18 observations. To avoid loss of observations and statistical power, I substitute the mean value of the percentage of the work force laid off (9.9%) for these missing observations. The coefficients on the multiple layoff dummy, the percentage of the work force laid off variable, and its squared term are insignificant, providing no support for Hypotheses 4a, 4b, or 5. These results must be interpreted with caution, since the sample size is quite small.

To determine whether collinearity is a problem in this regression, I also compute correlation coefficients for the independent variables in the Japanese sample. These correlation coefficients are reported in Table 6. With the exception of the percentage of the work force laid off and its squared term, the correlations are generally not large. If the regression is estimated without the squared term, the coefficient percentage of the work force laid off is still insignificant, suggesting that collinearity is not causing econometric problems.

## DISCUSSION

Consistent with previous studies and agency theory arguments, results show that layoff announcements trigger negative returns in both the United States and Japan. Interestingly, the stock price reactions of the Japanese sample are much smaller than the U.S. sample. One explanation for this difference is that the market index used in the computation of the abnormal returns in the Japanese sample may be problematic. Specifically, market index data are only available for the Nikkei index, which represents highly capitalized and large stocks. On the other hand, the U.S. sample uses a market value weighted index of *all* stocks in computing the abnormal returns.

A more convincing reason, however, is support for Hypothesis 2b, or evidence for the cross-holdings aspect of Aoki's cooperative game theory. Findings suggest that the effect of patient investors may outweigh the negative signal sent to investors when Japanese firms violate the tradition of lifetime employment and resort to layoffs as a way of coping with the firm's condition.

Nonetheless, results also show that there may be a limit to how patient investors actually are. Specifically, the Japanese results regarding single or multiple announcements are in contrast to the U.S. results, which reveal more negative returns for single announcements than for multiple announcements. Japanese investors may be willing to 'wait out' the period of financial distress

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Table 6. Correlations of independent variables for Japanese firms

| Variables | Permanent or temporary | Multiple or single event | Percentage of work force | (Percentage of work force)$^2$ |
|---|---|---|---|---|
| Permanent or temporary | 1.00 | | | |
| Multiple or single event | 0.20*** | 1.00 | | |
| Percentage of work force | –0.18 | –0.16** | 1.00 | |
| (Percentage of work force)$^2$ | –0.18 | 0.04 | 0.96*** | 1.00 |

***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

revealed by a single announcement. Multiple announcements, however, may send a signal that firms are experiencing severe problems and have been unable to adequately adjust to the competitive environment. In other words, multiple announcements establish the seriousness of the problems faced by the firm and cause investors to react accordingly.

The U.S. sample also reveals interesting differences based on layoff characteristics. For example, results show that permanent layoffs trigger much more negative returns than temporary layoffs. Temporary layoffs may not carry significant information about the expected change in performance of the sample firms. This is consistent with the inventory accumulation hypothesis of Saffer (1983) or seasonal demand shifts. There may be several possible interpretations to the negative returns for permanent layoffs. Possibly, the market views these layoff announcements as revealing problems regarding the future performance of the firm. Another possibility is that the market regards the layoffs as an insufficient remedial measure for firms in financial distress. Regardless, the evidence suggests that permanent layoff announcements are associated with significantly negative stock price reactions.

U.S. results also yield interesting results regarding the size of the layoff. Results suggest that perhaps investors perceive an optimal layoff size. Firms laying off smaller portions of their work forces, those in the lower quintiles, may have been perceived as not laying off 'enough'. On the other hand, firms dismissing the largest portion of their work force may be those that are having operational or financial problems. In such cases, the layoff announcements only verify the firm's conditions, making future hirings more difficult. These explanations may account for the curvilinear relationship between layoff size and abnormal returns. The size hypothesis has also been examined as a continuous independent variable in the multivariate regression of U.S. firms. Regression results show that, in general, stock prices react more negatively if layoffs affect a greater percentage of the firm's work force, and that the relationship may be curvilinear.

Even more interesting are the disparate returns triggered by proactive and reactive layoffs. Reactive layoff announcements yield significantly more negative returns than proactive layoff announcements. This finding is consistent with previous studies (Worrell *et al*., 1991; Cameron, 1994) and supports the notion that work force reductions alone do not necessarily provide the answers to corporate problems. Rather, restructuring and other strategic measures need to be implemented in conjunction with layoffs to correct problems of strategic competitiveness (Hoskisson and Hitt, 1994; Hamel and Prahalad, 1994).

The U.S. results also suggest that the sequencing of announcements affects the manner in which investors perceive layoffs in the United States. Specifically, results show that abnormal returns to the first layoff announcement in a given industry are −3.59 percent, while subsequent layoffs are significantly less negative (−0.97). This indicates that the subsequent layoffs may be perceived as nothing more than a mimetic reaction to an industry effect. Similarly, this change in attitude has also been accompanied by media portrayal from corporate downsizings to 'rightsizings,' supporting the notion that layoffs may have become 'institutionalized' as a popular method for coping with the competitive environment. However, findings also reveal that laying off workers alone is not a panacea for all corporate

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problems; firms need to employ strategies in conjunction with layoffs to proactively cope with the environment.

In Japan, stock price changes do not depend on layoff characteristics. One explanation for this is that the leakage of information makes the stock prices changes hard to discover. Large shareholders in Japan may have prior information pertaining to the layoff and may consequently respond before the layoff announcement. To assess whether there is any leakage, cumulative abnormal returns are calculated over −30, −2 and −15, −2 windows. These returns are not significantly different from zero, and imply that there is no evidence of leakage prior to the announcement.[9] A more convincing explanation is that the nature of Japanese firms is such that layoff characteristics do not affect the stock prices the way they do in the United States. This explanation is consistent with Kang and Stulz (1996), who argue that there is something fundamentally different about Japanese corporate governance.

## CONCLUSION

This study attempts to provide a comprehensive analysis of the differing impacts of layoff announcements on stock prices in the United States and Japan. In doing so, it begins with the premise that U.S. and Japanese firms resolve agency problems differently. Specifically, agency theory, and Aoki's cooperative game theory reveal fundamental differences in the organizational structure and contracting arrangements of U.S. and Japanese firms. These differences suggest that layoff announcements in the United States and Japan will have disparate impacts on stock prices. Indeed, empirical evidence uncovers four main findings.

First, evidence indicates that stock price reactions of layoff announcements in the United States differ from those in Japan. These disparities emphasize the potentially different effects of national culture on market structure, organizational form, and the effectiveness of differing strategies. Furthermore, these results accentuate the need for more cross-national studies.

Second, layoff announcements are associated with significantly negative returns in the United States. This suggests that investors may regard layoffs as a sign of financial distress, rather than an optimal way of responding to the competitive environment. In other words, layoffs may be a quick fix, rather than a strategic rethinking and rebuilding of core competencies to ensure a future growth. 'Getting smaller is not enough. Downsizing, the equivalent of corporate anorexia can make a company thinner; it doesn't necessarily make it healthier' (Hamel and Prahalad, 1994: 11). Firms should think twice before employing layoffs as a strategy.

Third, layoff announcements trigger a small (–0.56%) but negative reaction in Japan. The fact that the announcement effect for the Japanese firms is negative and significant also has several important consequences. It reveals that layoffs are a nontrivial economic event for Japanese firms. Japanese investors clearly view layoffs negatively. Thus, the frequently cited phenomenon of patient capital may be insufficent in preventing stock price declines.

Finally, stock price changes differ depending on the nature of the layoff. In the United States, layoffs seem to be a popular method of dealing with the economic environemt. The results indicate that layoffs are largely ineffective without strategies for coping with tomorrow's competitive environment (Hamel and Prahalad, 1994; Hoskisson and Hitt, 1994). Proactive layoffs are associated with less negative returns than reactive layoffs. Additionally, stock market reactions are more negative for single-event layoffs, for permanent layoffs, for firms laying off a greater percentage of the work force, and for layoff announcements in the early 1990s. These results spawn a profusion of research questions that have yet to be examined.

The results of this study also open avenues for more research questions on the effect of governance issues on downsizings. For example, Japanese results suggest that patient capital may not be quite as patient as one may expect. Future studies may examine just how patient Japanese shareholders can be. If Japanese shareholders are unlikely to see their stock, there should be little or no effect on stock market prices. Instead, the effects may be seen via corporate governance

[9] Kang and Stulz (1996) also discuss the possibility of leakage effects, albeit in a different context. They find no evidence of leakage. I also examine abnormal returns over the −30, −2 and −15, −2 windows for the U.S. sample and find no evidence of leakage.

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actions, such as pressure for removal of top management by large blockholders. Furthermore, the strong adherence to a tradition of lifetime employment has led several Japanese firms to strategic alternatives, such as hiring freezes, asset sales, and sales of divisions (Kang and Shivdasani, 1997). The effectiveness and aftermath of these strategies should also be further examined. For example, a breach of the lifetime employment contract may result in anger, decreased commitment, and stress for Japanese survivors of layoffs. Examining the effect of governance issues on different aspects of downsizings should enhance our understanding of their theoretical linkages.

In sum, future research should address several issues. First, research should better support or refute theoretical linkages to enhance our understanding of layoffs and downsizings. Second, research should explore the linkages between governance structures and other aspects of layoffs. Third, additional research on the effects of layoffs on financial performance should also enhance our understanding of the differing aspects of layoffs. Finally, results also suggest that other corporate strategies, such as diversification and joint ventures, may have differing performance implications across countries.

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